What we're doing

How well does the best contacts-v1 1.5B model @eric-czech trained in #61/#75 (eval loss 2.7566) predict residue–residue contacts on our curated eval set, alongside every other predictor — Protenix-v2 (single-seq / MSA), ESMFold, ESMFold2? This is exp82's named open follow-up: does careful tuning turn the near-chance #67 model into a real contact predictor?

The model and the eval

Model: W&B `prot-exp75-cv1-1_5b-e8-lr1e-3-wd0p2-v1` (Qwen3 1.47B; epochs 8, lr 1e-3, wd 0.2; eval/contacts-v1-val/loss = 2.756602 at step 35679), exported to HF and scored with the exp82 **pairwise** method (the best inference approach from #82). Eval: 554 proteins (FoldBench-100 + exp65 low-MSA / novel-fold), pyconfind side-chain ground-truth contacts, the same resolved-residue universe for every predictor. Metrics: AUC + precision @ {L, L/2, L/5, R}, by range.

Finding 1 — ranking AUC is competitive

MarinFold's long-range contact-ranking **AUC ≈ 0.88 (0.90 aggregate)** matches **ESMFold** and beats **single-sequence Protenix-v2** (0.81). It is **robust to MSA depth and fold novelty** — it even **edges ESMFold2 on novel folds** (0.81 vs 0.79). So careful tuning took the model from ~chance (the quick #67 model in exp82) to ESMFold-class ranking AUC, **from sequence alone**.

Finding 2 — top-K precision lags structure methods

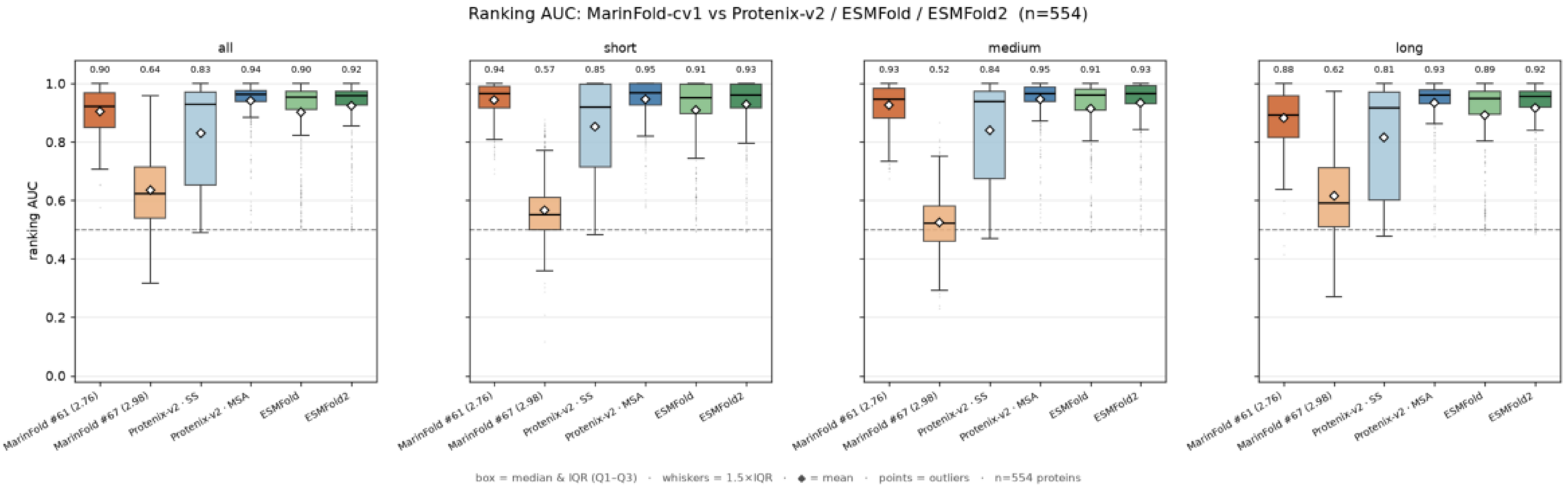
At the sharp end — R-precision and contacts@L — MarinFold (≈ 0.27 / 0.19 long-range) sits **well below every structure predictor** (ESMFold2 0.77 / 0.44; Protenix-MSA 0.80 / 0.47). The LM orders contacts broadly well but does not concentrate confidence on the true top contacts.

Takeaway

A 1.5B **sequence-only** LM now has a real, MSA-depth-robust contact signal (high AUC) but is not yet a high-precision contact predictor (low top-K precision). Note AUC somewhat favours a full continuous ranker (MarinFold scores every pair) over structure-derived contact sets (degree 0 for most pairs); the top-K precision metrics are the fairer "did you find the contacts" comparison, and there structure + MSA still lead. Plots: contacts @ {L, L/2, L/5}, R and AUC by predictor and range, stratified by MSA-depth tier and fold novelty.

contacts_at_AUC_by_config_and_range.png

ranking AUC vs pyconfind contacts (degree>=0.001, sep>=6), per predictor, aggregate and by range, over 554 proteins. Boxplots: box = median & IQR, whiskers = 1.5×IQR, points = outliers, white diamond = mean (labelled). MarinFold ranks pairs by its pairwise contact-statement log-prob; the structure models rank by pyconfind degree on the predicted structure.

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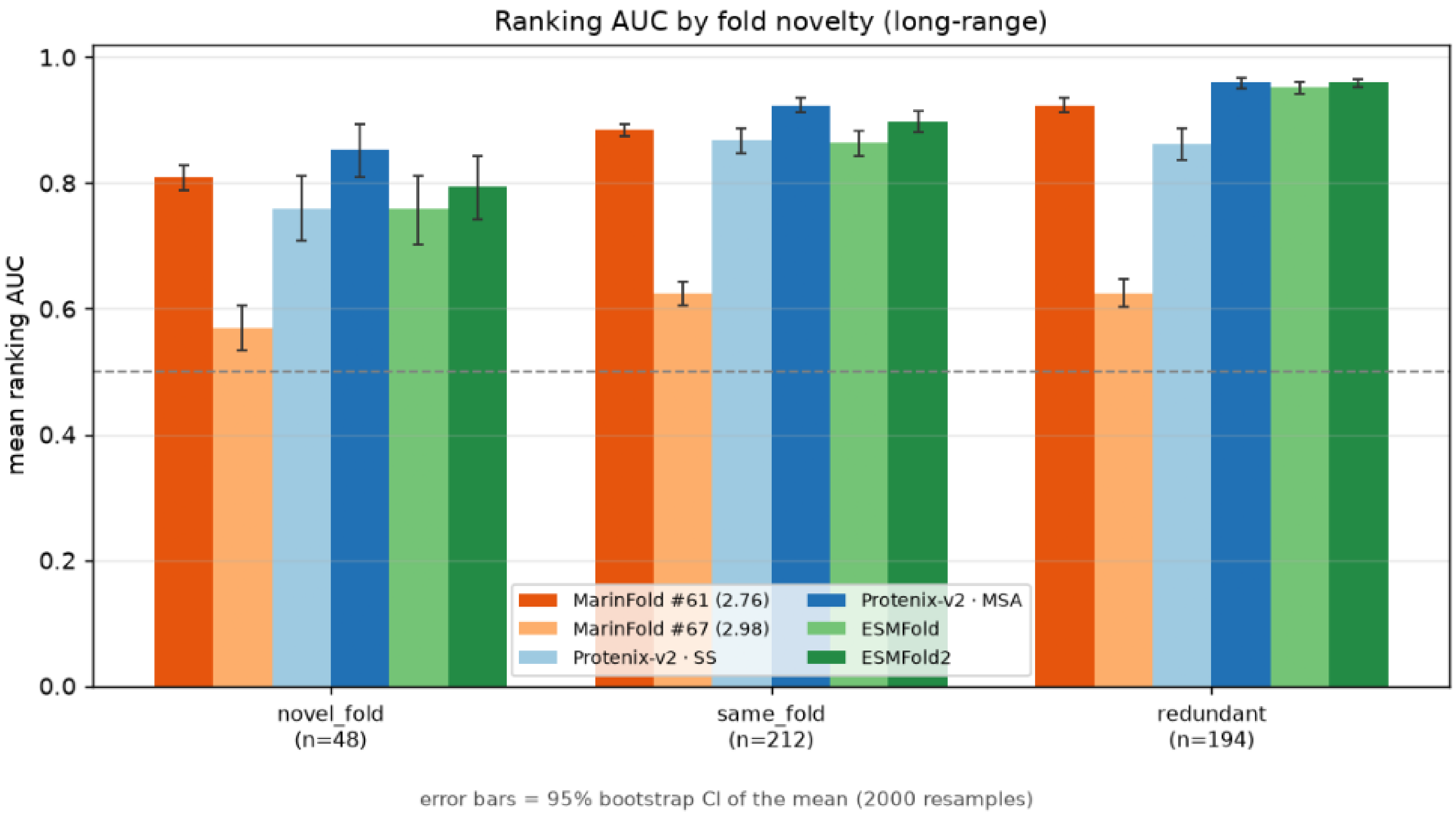
$ plot.py --precision-csv data/contact_precision_all.csv --out plots
```

contacts_at_AUC_by_fold_verdict.png

Ranking AUC by fold novelty (long-range). Bars = mean; error bars = 95% bootstrap CI of the mean.

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$ plot.py --precision-csv data/contact_precision_all.csv --out plots
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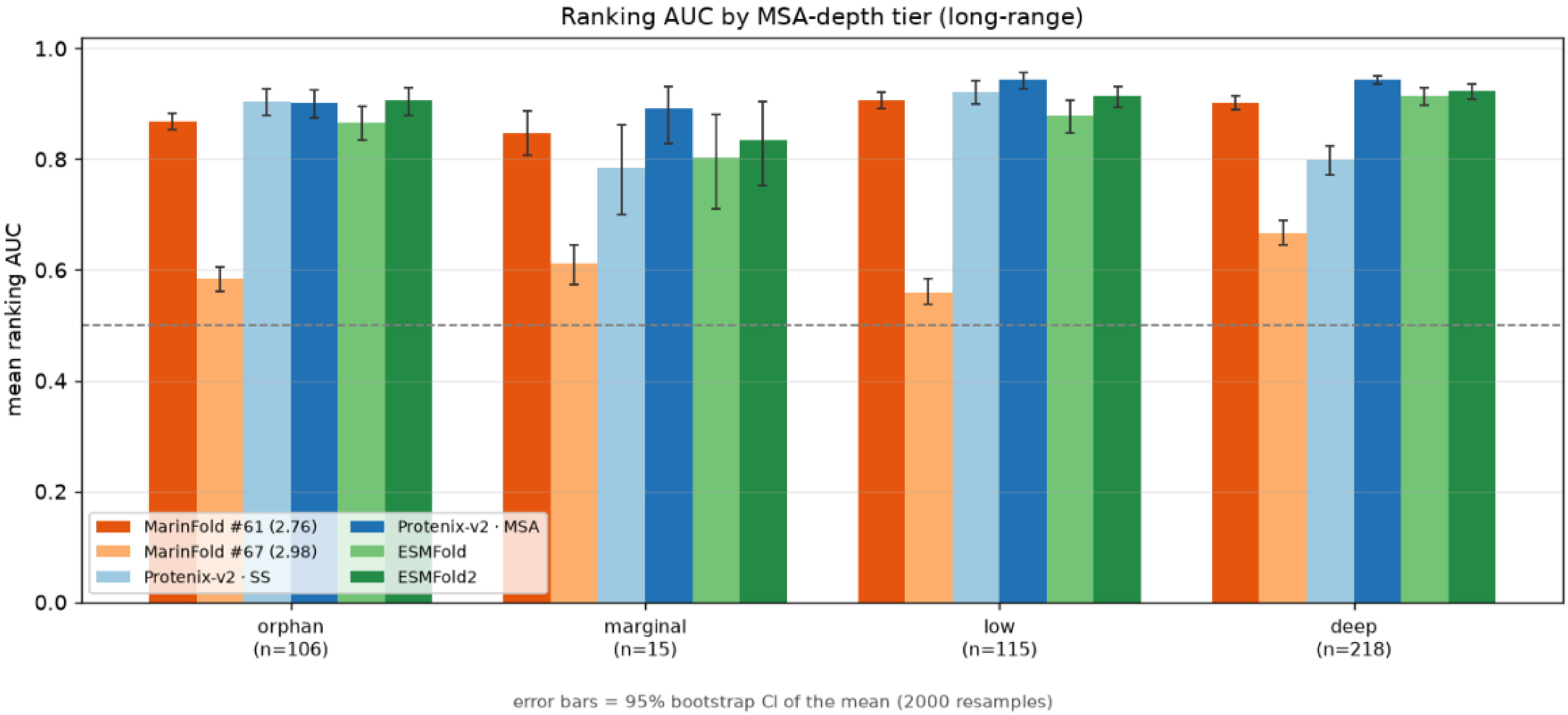
contacts_at_AUC_by_neff_tier.png

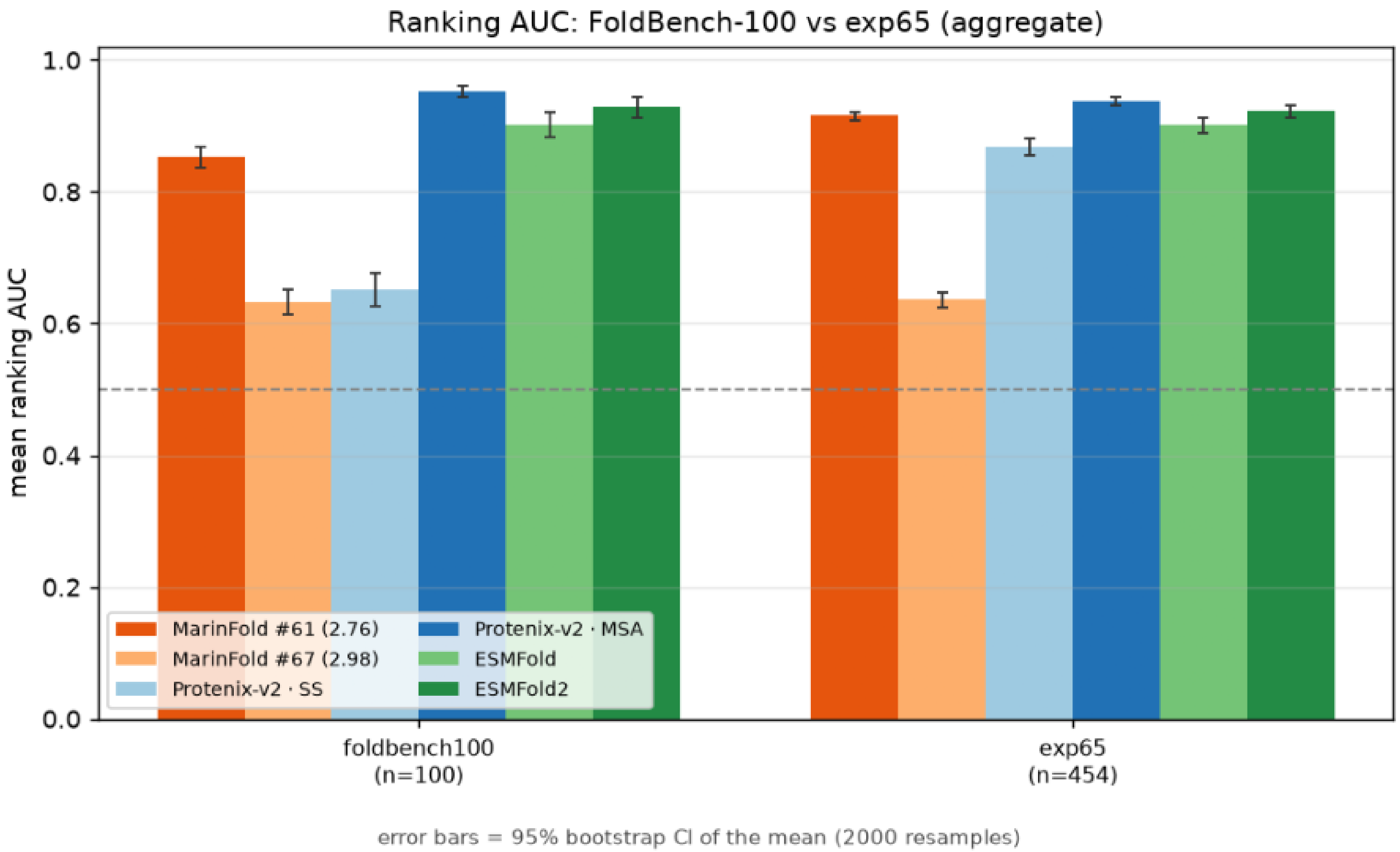
Ranking AUC by MSA-depth tier (long-range). Bars = mean; error bars = 95% bootstrap CI of the mean.



`$ plot.py --precision-csv data/contact_precision_all.csv --out plots`

contacts_at_AUC_foldbench_vs_exp65.png

Ranking AUC: FoldBench-100 vs exp65 (aggregate). Bars = mean; error bars = 95% bootstrap CI of the mean.

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$ plot.py --precision-csv data/contact_precision_all.csv --out plots
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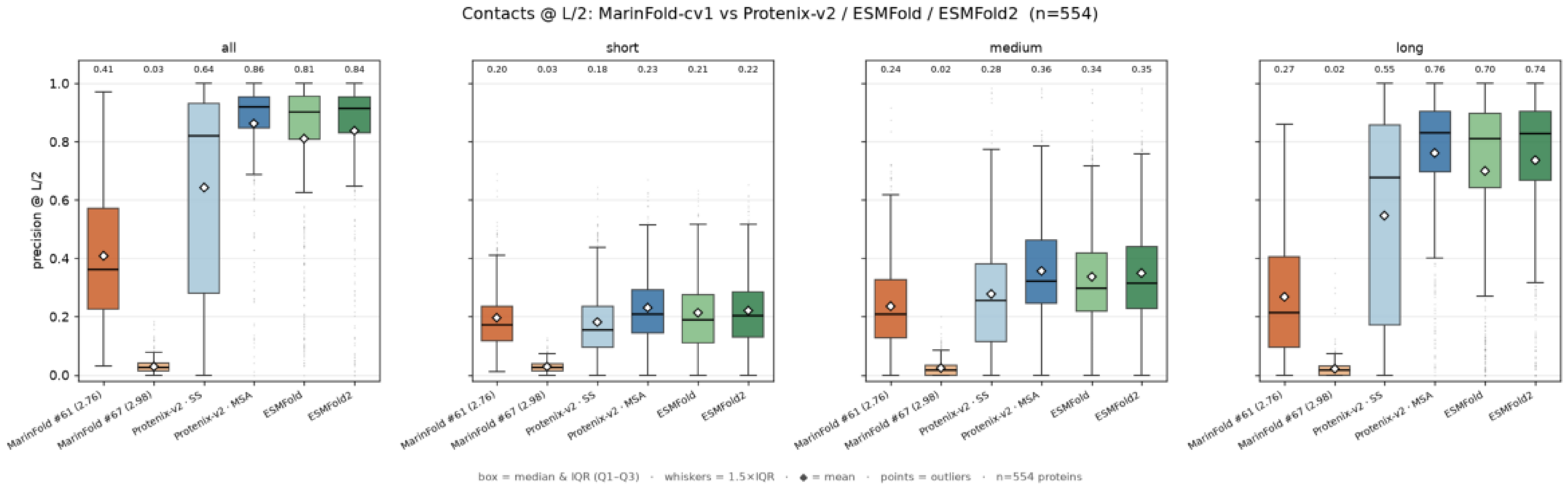
contacts_at_L_2_by_config_and_range.png

precision @ L/2 vs pyconfind contacts (degree>=0.001, sep>=6), per predictor, aggregate and by range, over 554 proteins. Boxplots: box = median & IQR, whiskers = 1.5×IQR, points = outliers, white diamond = mean (labelled). MarinFold ranks pairs by its pairwise contact-statement log-prob; the structure models rank by pyconfind degree on the predicted structure.

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$ plot.py --precision-csv data/contact_precision_all.csv --out plots
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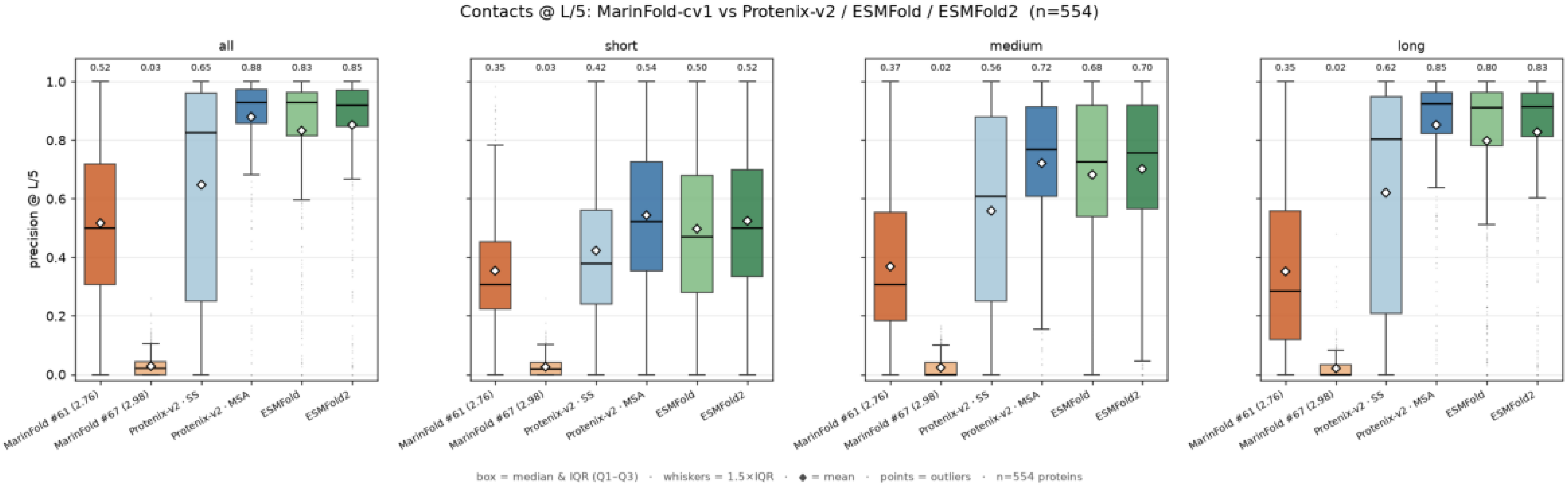
contacts_at_L_5_by_config_and_range.png

precision @ L/5 vs pyconfind contacts (degree>=0.001, sep>=6), per predictor, aggregate and by range, over 554 proteins. Boxplots: box = median & IQR, whiskers = 1.5×IQR, points = outliers, white diamond = mean (labelled). MarinFold ranks pairs by its pairwise contact-statement log-prob; the structure models rank by pyconfind degree on the predicted structure.

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$ plot.py --precision-csv data/contact_precision_all.csv --out plots
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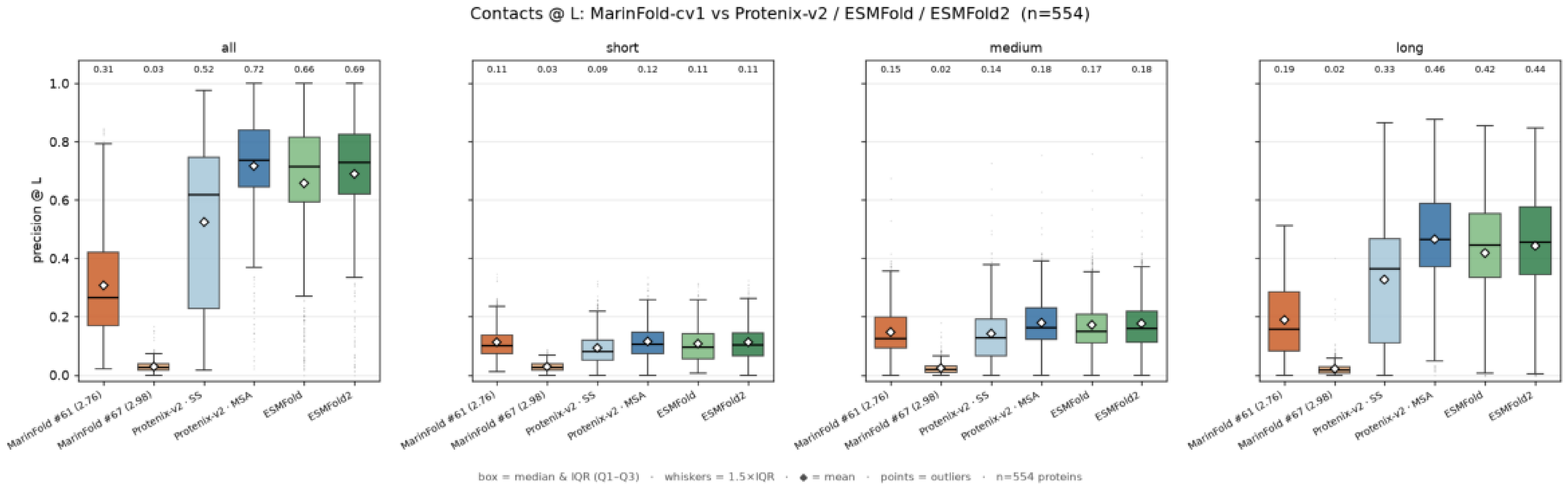
contacts_at_L_by_config_and_range.png

precision @ L vs pyconfind contacts (degree>=0.001, sep>=6), per predictor, aggregate and by range, over 554 proteins. Boxplots: box = median & IQR, whiskers = 1.5×IQR, points = outliers, white diamond = mean (labelled). MarinFold ranks pairs by its pairwise contact-statement log-prob; the structure models rank by pyconfind degree on the predicted structure.

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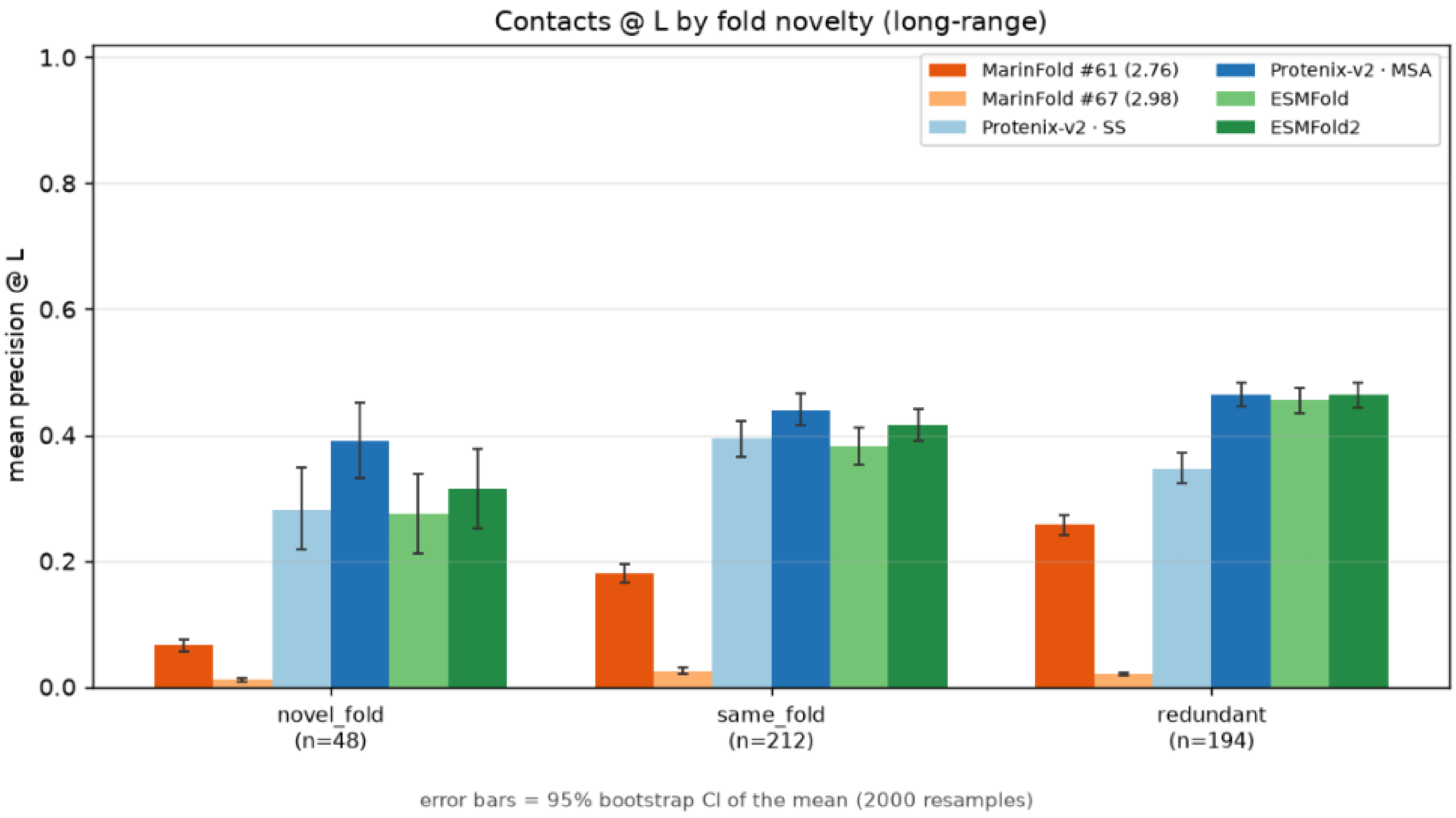
contacts_at_L_by_fold_verdict.png

Contacts @ L by fold novelty (long-range). Bars = mean; error bars = 95% bootstrap CI of the mean.

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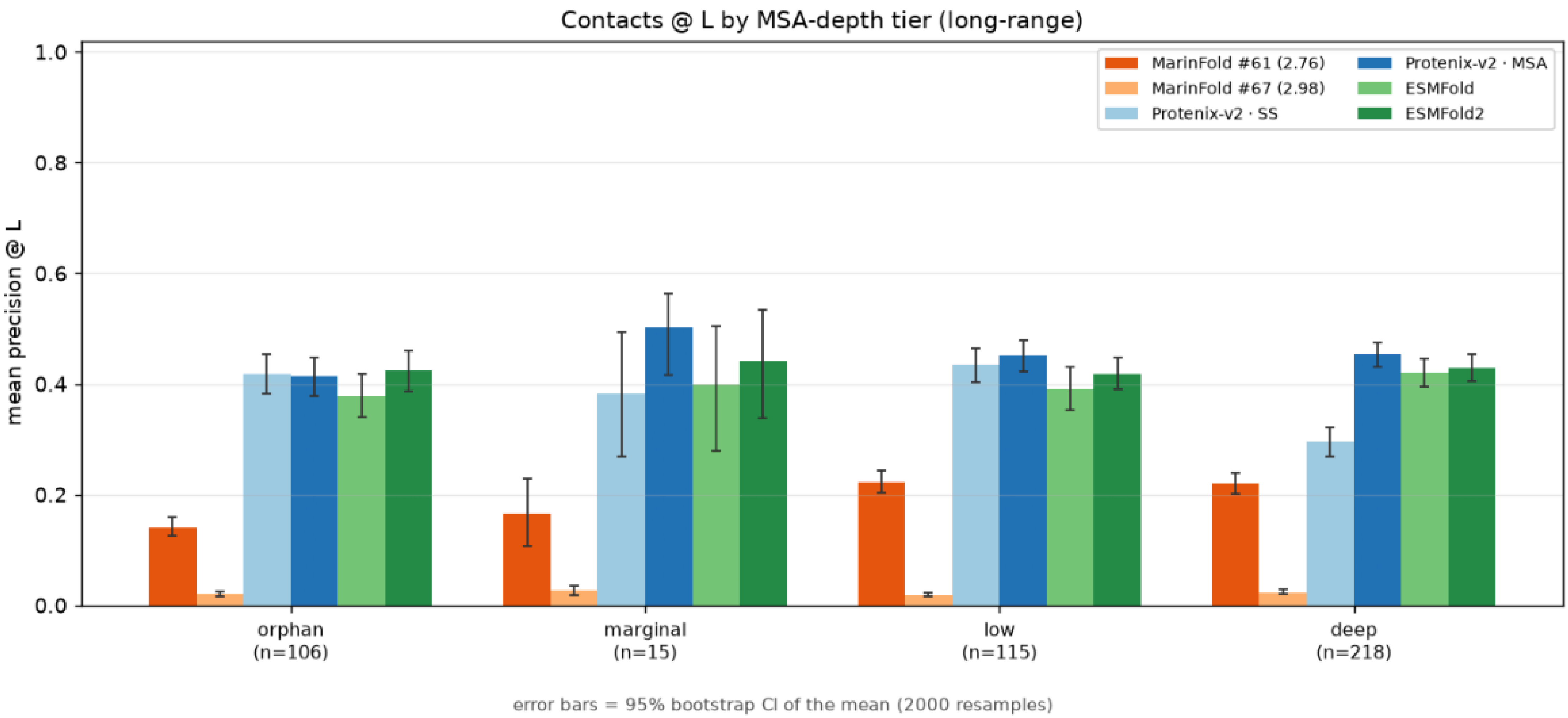
contacts_at_L_by_neff_tier.png

Contacts @ L by MSA-depth tier (long-range). Bars = mean; error bars = 95% bootstrap CI of the mean.

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$ plot.py --precision-csv data/contact_precision_all.csv --out plots
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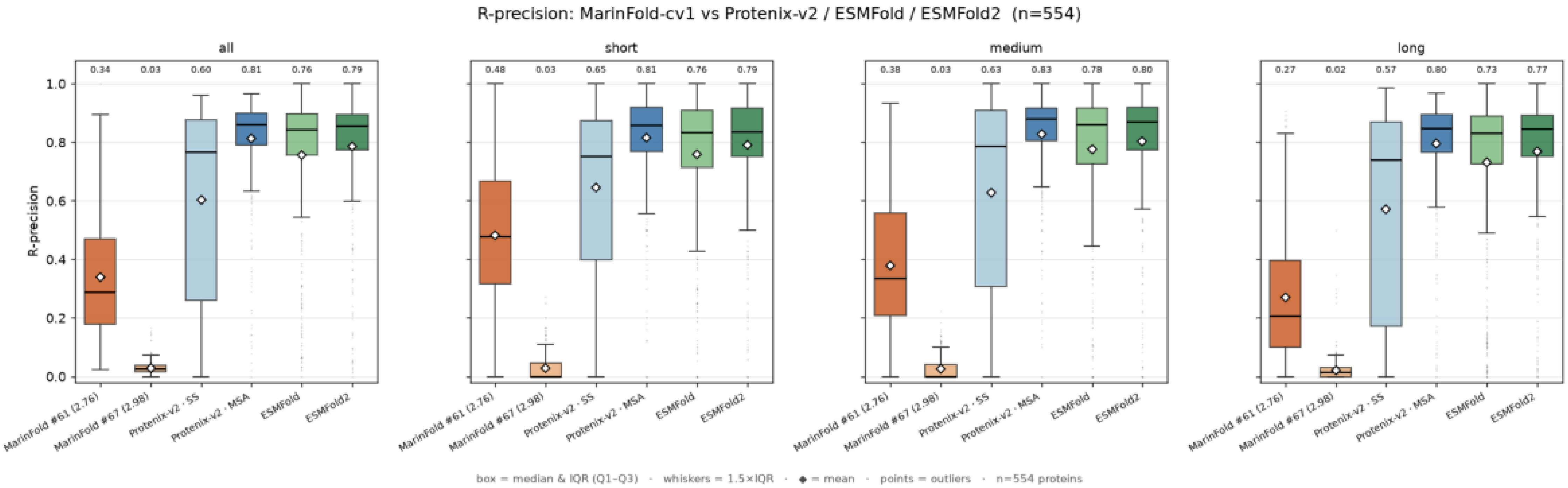
contacts_at_R_by_config_and_range.png

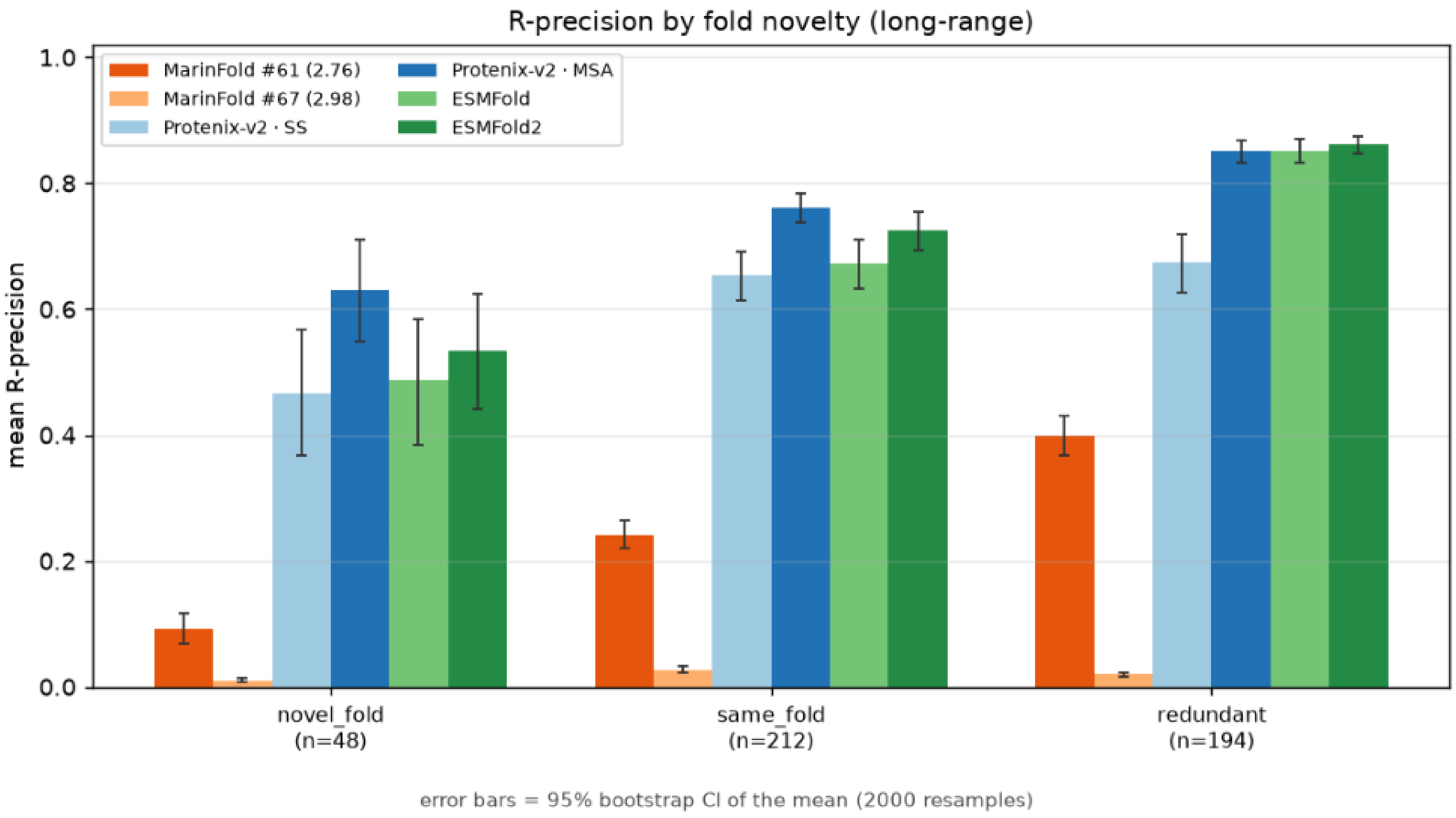
R-precision vs pyconfind contacts (degree>=0.001, sep>=6), per predictor, aggregate and by range, over 554 proteins. Boxplots: box = median & IQR, whiskers = 1.5×IQR, points = outliers, white diamond = mean (labelled). MarinFold ranks pairs by its pairwise contact-statement log-prob; the structure models rank by pyconfind degree on the predicted structure.



$ plot.py --precision-csv data/contact_precision_all.csv --out plots

contacts_at_R_by_fold_verdict.png

R-precision by fold novelty (long-range). Bars = mean; error bars = 95% bootstrap CI of the mean.

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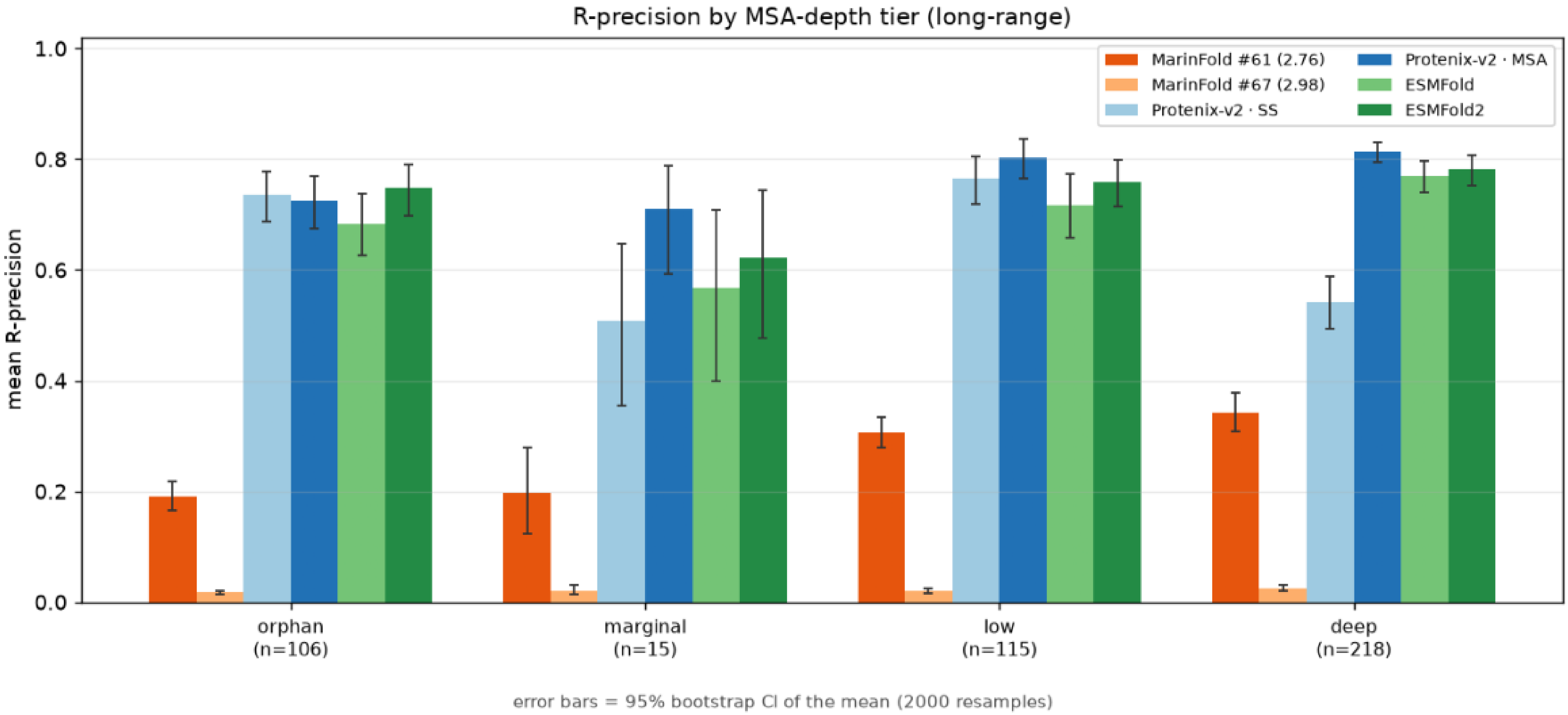
contacts_at_R_by_neff_tier.png

R-precision by MSA-depth tier (long-range). Bars = mean; error bars = 95% bootstrap CI of the mean.

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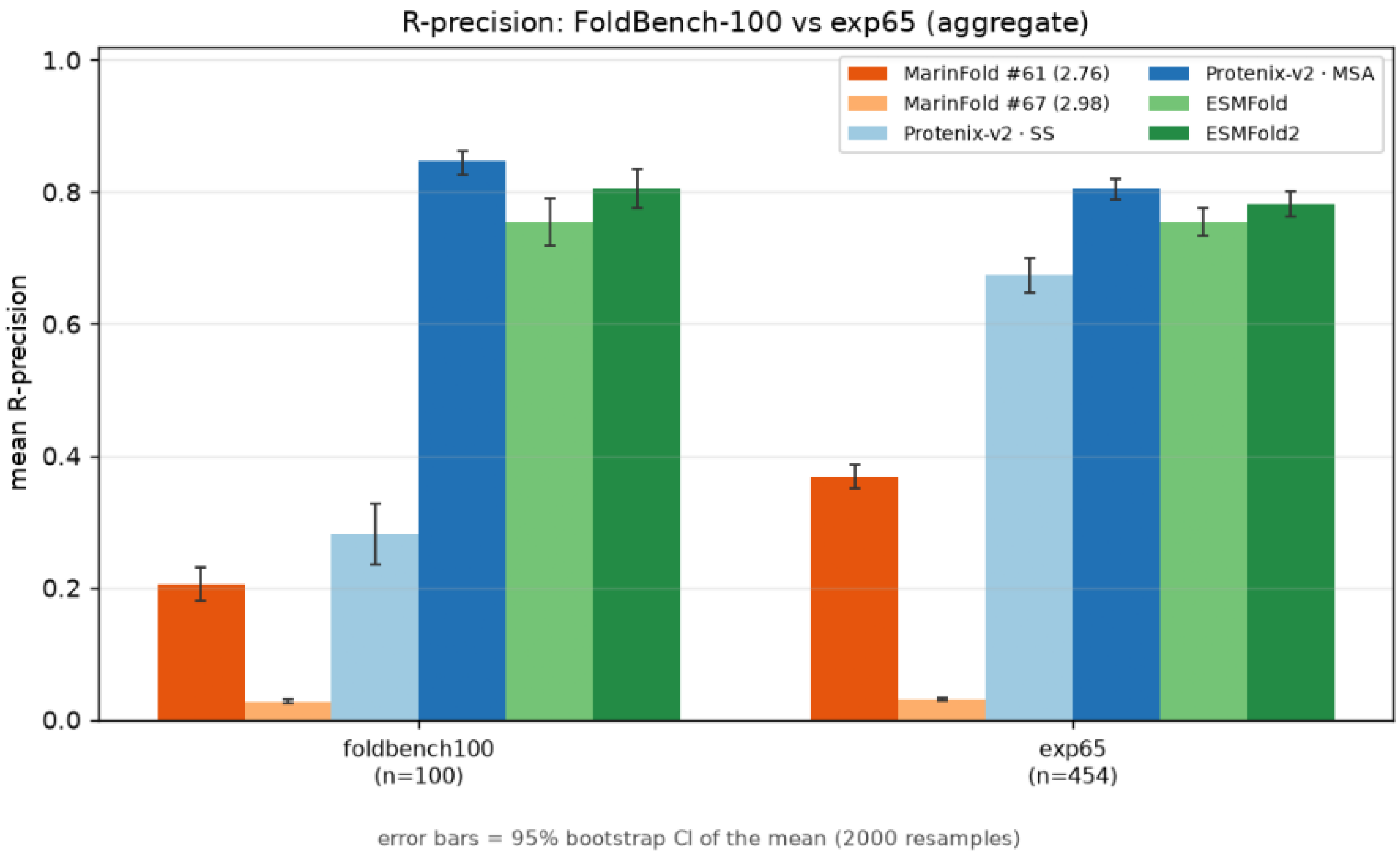
contacts_at_R_foldbench_vs_exp65.png

R-precision: FoldBench-100 vs exp65 (aggregate). Bars = mean; error bars = 95% bootstrap CI of the mean.

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$ plot.py --precision-csv data/contact_precision_all.csv --out plots
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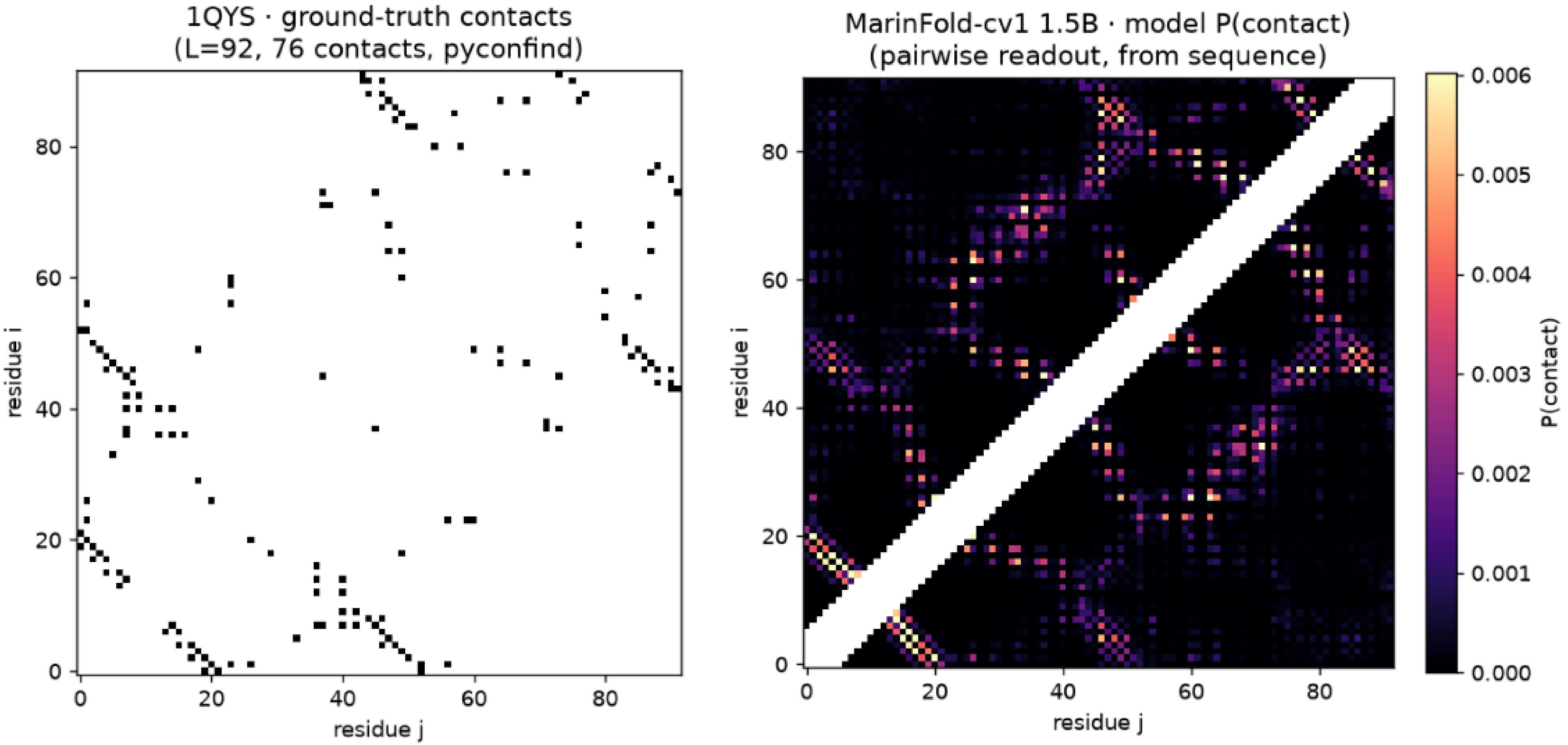
heatmap_1QYS.png

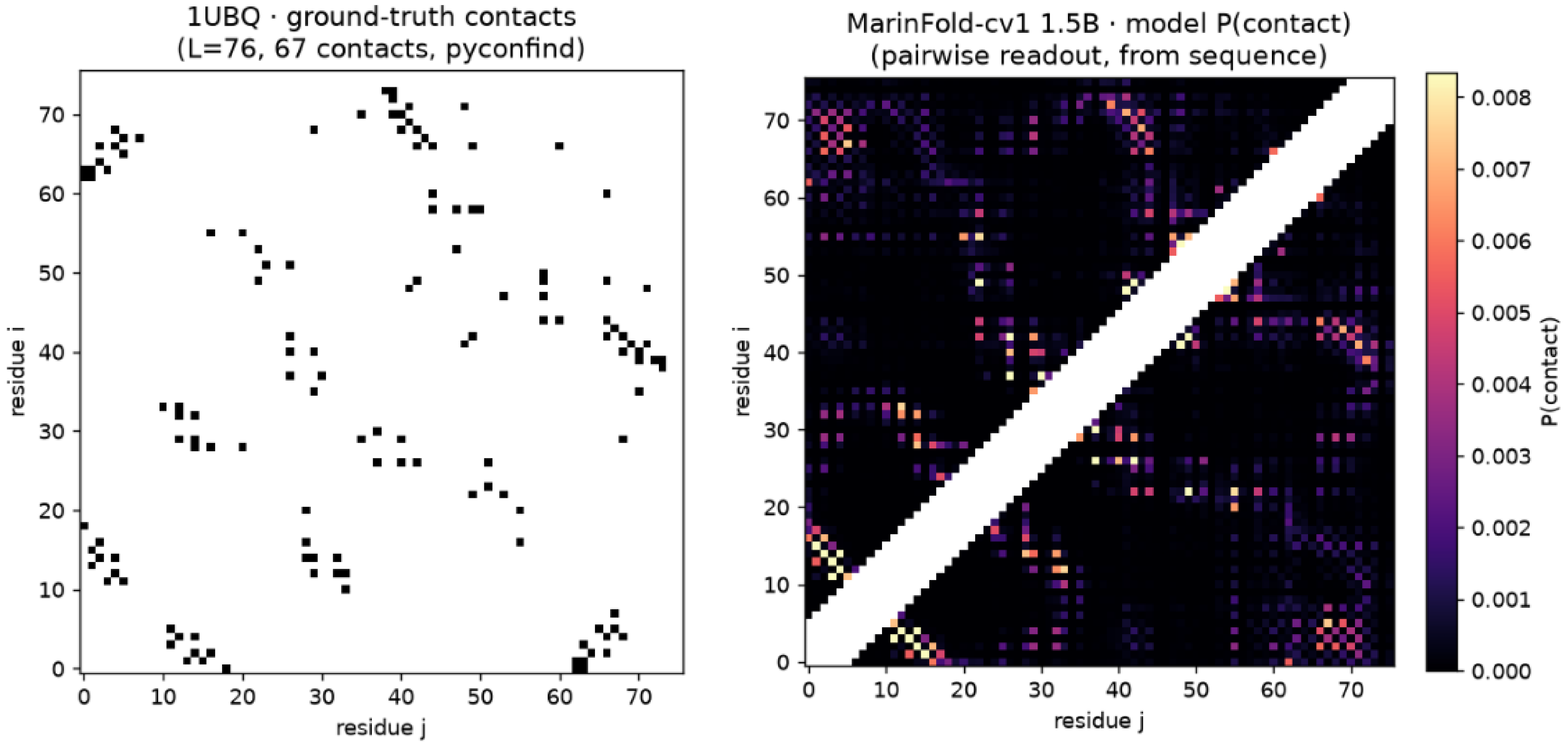
1QYS: ground-truth pyconfind contacts (left) vs the contacts-v1 1.5B model's P(contact) from sequence (right) — the probability the model emits each pair as its next contact statement. Near-diagonal band (sep<6) masked.



$ benchmark_heatmaps.py

heatmap_1UBQ.png

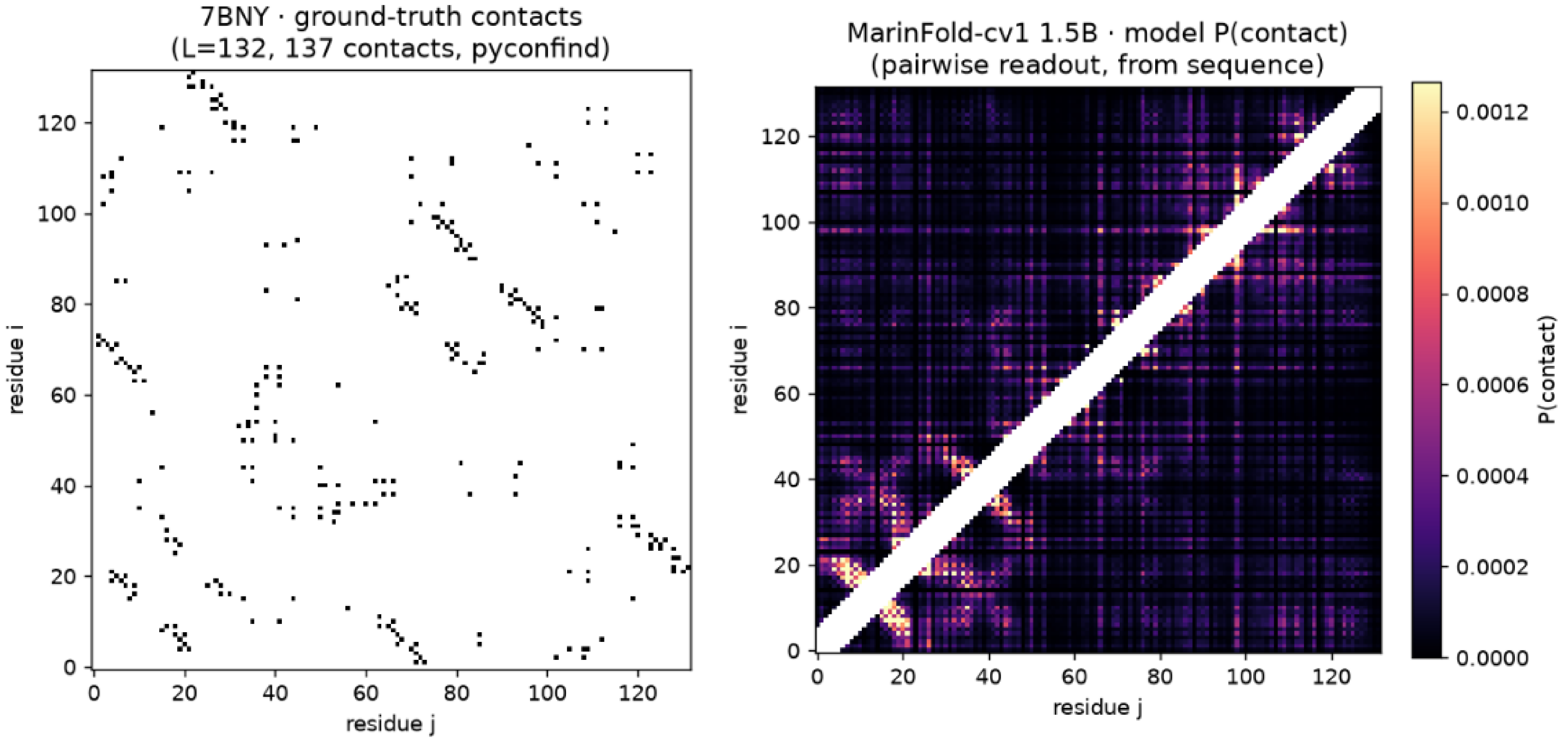
1UBQ: ground-truth pyconfind contacts (left) vs the contacts-v1 1.5B model's P(contact) from sequence (right) — the probability the model emits each pair as its next contact statement. Near-diagonal band (sep<6) masked.



$ benchmark_heatmaps.py

heatmap_7BNY.png

7BNY: ground-truth pyconfind contacts (left) vs the contacts-v1 1.5B model's P(contact) from sequence (right) — the probability the model emits each pair as its next contact statement. Near-diagonal band (sep<6) masked.



$ benchmark_heatmaps.py